

BrailleGuide: A Vision-Language-Model-Powered Assistive System for Environmental Understanding of Visually Impaired Users

Yuxiang Chen

High School Student

Beijing 101 High School International Department

Beijing, China

c19801354136@outlook.com

Abstract—Visual impairment can significantly limit independent environmental perception, while many existing assistive systems provide only limited information from visual input. Conventional systems often depend on object detection or Optical Character Recognition (OCR), which may identify individual objects or text but cannot always explain scene context, user-relevant relationships, or potential actions. Vision-Language Models (VLMs) offer a promising alternative by jointly interpreting visual and linguistic information. This paper proposes BrailleGuide, a split-form wearable assistive system that connects a VLM-based Visual Perceptor unit with a Handheld Terminal containing a refreshable Braille display, buttons, and a joystick. The perception unit captures environmental images, sends them to an online VLM, and converts the generated descriptions into tactile feedback. The main contributions are a VLM-based accessibility pipeline for environmental description, a lightweight split architecture separating perception from tactile interaction, and a prototype evaluation that demonstrates the feasibility of VLM-assisted environmental awareness through Braille feedback.

Index Terms—vision-language model, assistive system, visually impaired users, environmental understanding, tactile Braille display

I. INTRODUCTION

Visual information access remains a major challenge for people with visual impairments (PVis). Although assistive technologies have improved accessibility, PVis may still encounter difficulties obtaining timely and situated information about unfamiliar environments [1], including objects, signs, spatial layouts, and potential safety risks.

Computer vision has been widely explored to reduce these barriers. Recent systems such as *AIDEN* combine object detection, OCR, and LLaVA-based scene understanding for object recognition, reading, and environmental question answering [2]. However, many assistive systems still emphasize object-level recognition or audio-based feedback.

Vision-Language Models (VLMs) provide new opportunities because they can generate natural-language descriptions from image inputs and user prompts [3]. For PVis, however, the output channel is also important. Audio feedback may be unsuitable in noisy, quiet-required, or privacy-sensitive settings, while refreshable Braille displays offer a private

and reviewable tactile method that remains less commonly integrated with VLM-based environmental understanding.

In this work, BrailleGuide, a VLM-powered multimodal assistive system that connects camera-based visual perception with a tactile Braille output device is proposed. The system adopts a split architecture with a visual perceptor that captures images and communicates with an online VLM, while a handheld terminal provides user input and refreshable Braille output. Users can actively request environmental information and receive semantic descriptions of different directions through the tactile interface.

The main contributions of this work are as follows:

- A VLM-based perception pipeline that transforms environmental images into concise descriptions suitable for Braille feedback.
- A lightweight split system architecture that separates visual perception and user interaction into a camera-based perception unit and a handheld Braille terminal.

II. BACKGROUND

A. Vision-Based Assistive Systems

Vision-based assistive systems commonly use object detection, OCR, image captioning, or visual question answering to convert camera inputs into accessible information. VizWiz highlighted the practical need for question-driven visual assistance by collecting images and questions from blind users in real-world situations [4]. More recent systems integrate multiple perception modules; for example, *AIDEN* combines YOLO, OCR, and a large vision-language assistant to support environmental descriptions and user questions [2]. These systems motivate BrailleGuide’s perception task, but their interaction channels are still mainly auditory.

B. Vision-Language Models

VLMs extend earlier vision pipelines by aligning image representations with language models, enabling open-ended description, reasoning, and question answering. GPT-4V demonstrates the capability of multi-modal models to process image inputs and produce textual responses [3], while LLaVA shows that visual instruction tuning can create a practical large

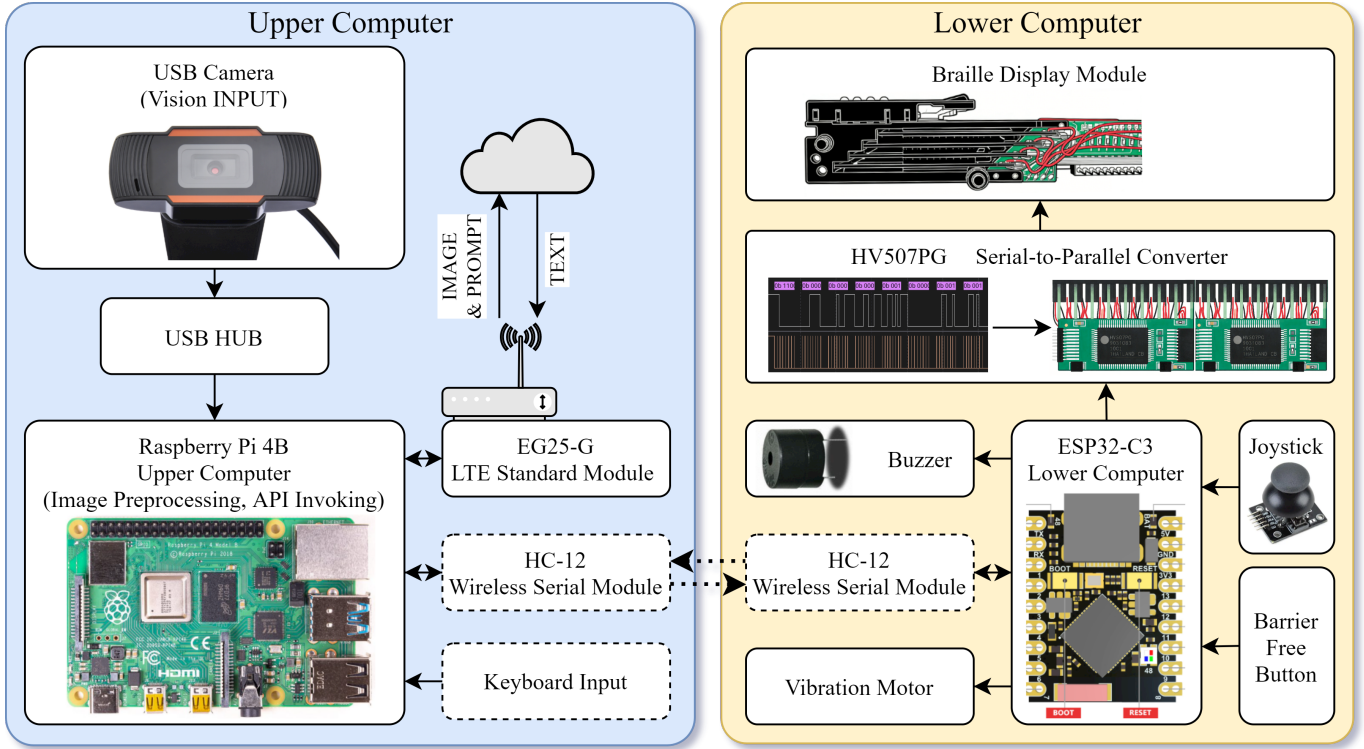


Fig. 1. Hardware Design of BrailleGuide

multimodal assistant from a vision encoder and a language model [5]. For assistive use, this ability is valuable because users may ask context-dependent questions that cannot be answered by a fixed detector alone.

C. Tactile and Braille Interfaces

Refreshable Braille displays provide dynamic tactile access to digital text by updating Braille cells under electronic control. Recent surveys describe their importance for nonvisual information access [6]. In the context of environmental understanding, Braille output can complement speech by providing private, persistent, and rereadable descriptions, which supports the motivation for connecting VLM-generated text with a handheld tactile terminal. However, few existing systems integrate VLMs with Braille feedback, leaving a gap that BrailleGuide aims to address.

III. BRAILLEGUIDE SYSTEM ARCHITECTURE

A. Overall Architecture

BrailleGuide adopts a split architecture composed of two cooperative modules: a VLM-based Visual Perceptor and a Handheld Braille Terminal. The Visual Perceptor functions as the upper computer and is responsible for camera-based image acquisition, online VLM communication, response parsing, and Braille-code generation. The Handheld Braille Terminal functions as the lower computer and provides direct user interaction through buttons, a joystick, vibration feedback, and a refreshable Braille display. This separation allows the

computation-heavy and network-dependent tasks to remain on the perception unit, while the user-facing terminal can stay compact and tactile-focused.

The end-to-end workflow is command driven. A user input from the handheld terminal is transmitted to the Visual Perceptor as a serial command, such as a capture request or a request for a specific directional field. The Visual Perceptor captures or reuses an image, sends the image and prompt to the online VLM, parses the structured response into environmental fields, converts the selected text into Braille tokens, and sends paginated Braille data back to the terminal. The terminal then decodes the received tokens and updates the Braille display. In this way, BrailleGuide connects visual perception, semantic scene understanding, and tactile reading in a single interaction loop.

B. Hardware Design

As shown in Fig. 1, the Visual Perceptor serves as the upper computer of BrailleGuide. Its core controller is a Raspberry Pi 4B, which is connected to a USB camera, an EG25-G LTE module, and an HC-12 wireless serial module. The USB camera captures environmental images from the user's surroundings. The EG25-G LTE module provides Internet access for online VLM requests, while the HC-12 module establishes the local wireless serial link between the Visual Perceptor and the Handheld Braille Terminal. In this design, the upper computer is responsible for image acquisition, network communication,

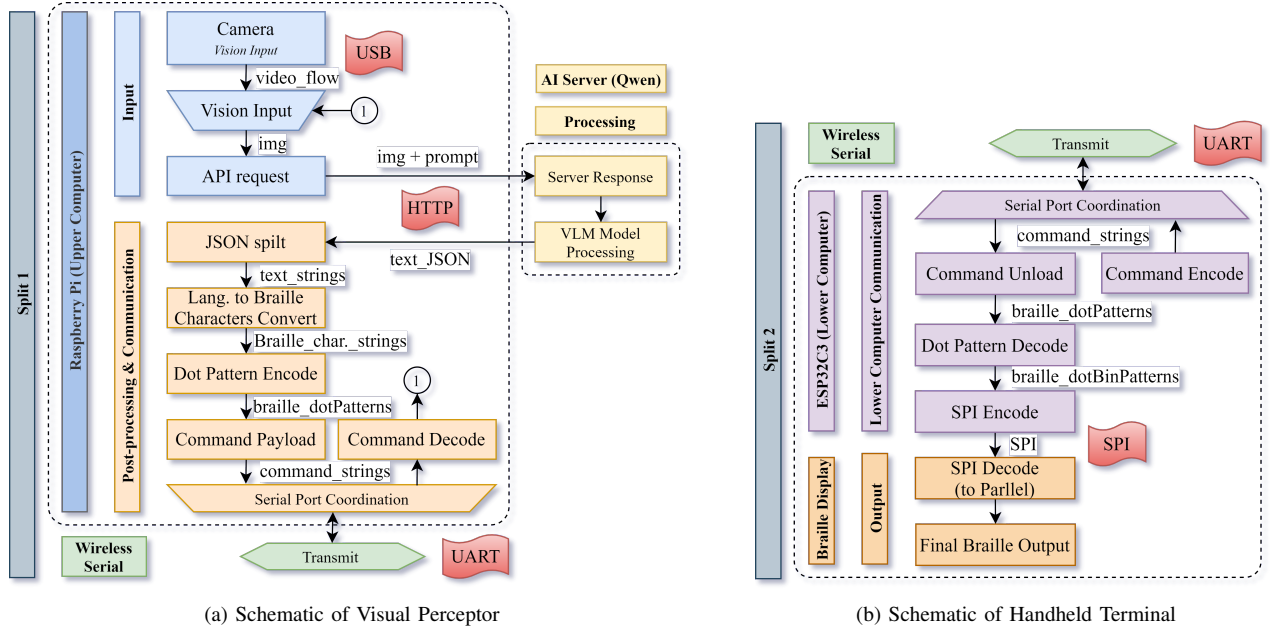


Fig. 2. Software data flow of BrailleGuide across the Visual Perceptor and Handheld Braille Terminal

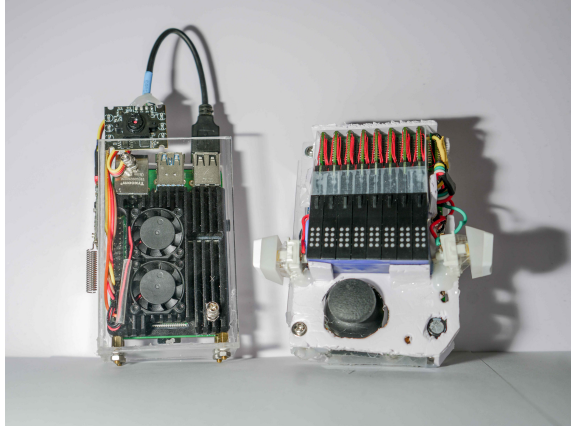


Fig. 3. BrailleGuide prototype with Visual Perceptor (left) and Handheld Braille Terminal (right)

VLM request handling, and transmission of processed Braille codes to the lower computer.

The Handheld Braille Terminal serves as the lower computer and user interaction unit. Its core controller is an ESP32-C3 Super Mini development board, which is connected to an 8-cell Braille display, a vibration motor, two barrier-free buttons, an XY joystick, and an optional buzzer. The ESP32-C3 receives Braille codes from the upper computer through the HC-12 wireless serial connection, decodes the received data, and drives the Braille display through parallel output. The buttons and joystick provide command input, such as requesting a new environmental description or selecting an interaction mode, while the vibration motor and buzzer can provide lightweight status notifications. Through this split hardware connection, computation, networking, and image capture remain on the

Visual Perceptor, while tactile output and direct user control remain on the handheld terminal.

C. Software Design

The Visual Perceptor runs Python on the Raspberry Pi, while the Handheld Braille Terminal runs MicroPython on the ESP32-C3. They communicate through a line-based UART protocol over the HC-12 link at 9600 baud. The Visual Perceptor handles camera input, serial commands, and online VLM inference by sending the current image as a base64 data URI with the selected prompt. The VLM returns structured JSON containing an overall description, five directional fields, and a text-information field.

The upper computer parses and stores the latest fields, then converts selected text into Braille tokens using a Chinese Braille translation module [7]. Since the display has eight cells, each field has an independent paging state and is sent one page at a time through messages such as `DATA BR <tokens>`. Status and errors use `EVT` and `ERR` messages, so the terminal can distinguish ordinary display data from waiting or fault conditions.

The terminal continuously polls its buttons, joystick, and UART interface. A joystick-button click sends `CAP` for a new capture, a double click requests text information, the two buttons send `BR PREV` and `BR NEXT`, and joystick directions request cached fields such as `BR FRN`, `BR LFT`, `BR RGT`, and `BR GRD`. When a `DATA BR` payload arrives, the ESP32-C3 converts its eight tokens into a 64-bit display state and updates the Braille cells through SPI with latch control. Vibration and blinking provide lightweight feedback for waiting, reception, and errors, while cached fields can be revisited without a new VLM request.

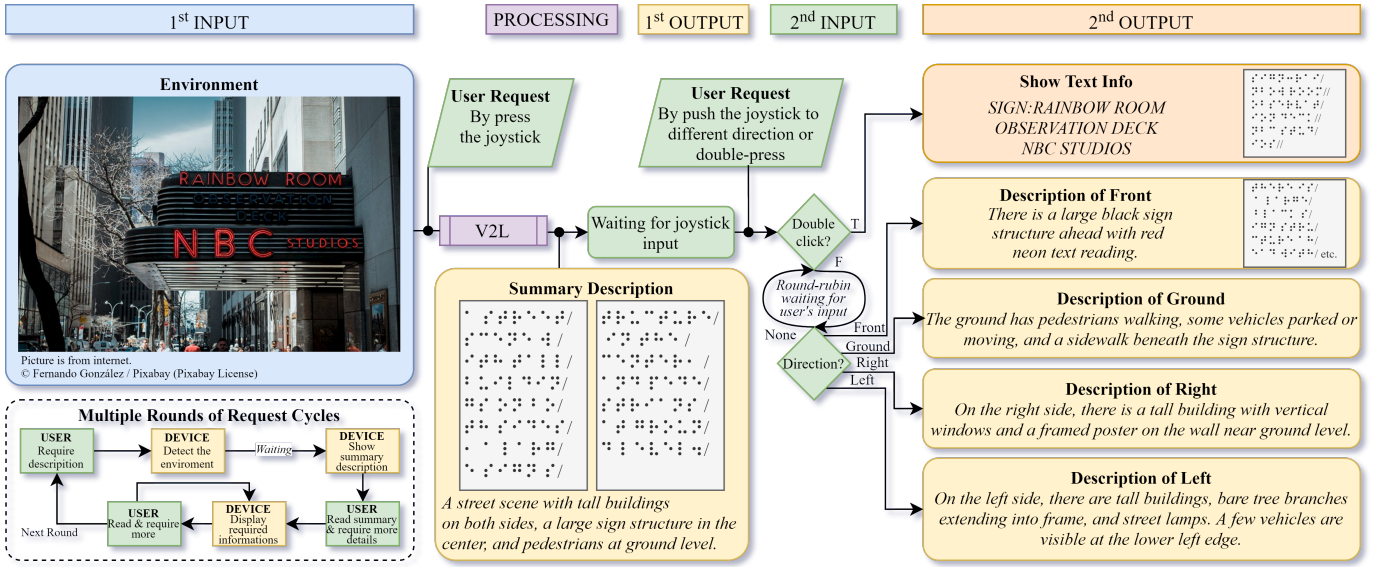


Fig. 4. Illustrative workflow, not additional participant-study evidence.

The command-driven design separates new recognition from subsequent field browsing, avoiding repeated image transmission and network calls.

IV. VLM-BASED PERCEPTION PIPELINE

A. Prompt Engineering

The prompt guides the VLM toward task-oriented environmental understanding for PVIs. It prioritizes passages, doors, stairs, ramps, vehicles, pedestrians, and obstacles, while requiring directly confirmed visual content.

Table I summarizes representative instructions that constrain the VLM to produce structured fields.

The fields match the handheld interaction: the joystick requests front, left, right, or ground; overall gives a short summary; and text_info stores readable signs. With only eight cells, decorative details and vague expressions are discouraged to simplify paging.

B. Semantic Response Processing and Braille Feedback

As shown in Fig. 4, the upper-computer program parses each JSON response, normalizes field names, and stores the latest result. This separates the VLM response format from the

terminal commands and allows the same result to be reused for several user requests. Nonempty fields are converted into Braille tokens, while malformed or unavailable fields generate an error instead of being displayed as ordinary content.

The terminal displays overall first, divides longer fields into eight-cell pages, and maintains independent page navigation for each selected field. The buttons move between pages, while the joystick requests cached directional fields without new recognition. Vibration and display feedback indicate waiting, successful reception, and error states, helping the user distinguish new output from an unavailable or incomplete response.

V. IMPLEMENTATION DETAILS

Some technical parameters are summarized in Table II. The VLM model can be replaced with any supported model that accepts the same prompt and returns the same JSON structure. The Braille translation module uses UEB Grade 1, which is widely supported and avoids ambiguity. The UART link uses a simple line-based protocol with CRLF termination, allowing easy parsing on both ends.

TABLE I
REPRESENTATIVE PROMPT INSTRUCTIONS

Component	Excerpted instruction
Role	Convert image into environmental descriptions for PVI.
Objective	Only output information that provides value for spatial understanding and action decision-making.
Source rule	Describe only content directly confirmed from the image; do not make assumptions or add uncertain information.
Priority	Emphasize safety-related and operational items.
Output	Return only a JSON object with overall, {front, left, right, ground, far}, text_info.

TABLE II
IMPLEMENTATION PARAMETERS

Parameter	Value
VLM model	qwen3-vl-flash or other supported model with same transmission protocol
Image preparation	720 px max.; JPEG quality 70
API timeout	60 s
UART link	HC-12; 9600 baud; CRLF
Braille Version	UEB Grade 1*
Braille Paging	8 cells/page
Input timing	5 ms polling; 80 ms button debounce

* “UEB Grade 1” means “Unified English Braille without abbreviations”.

VI. EVALUATION

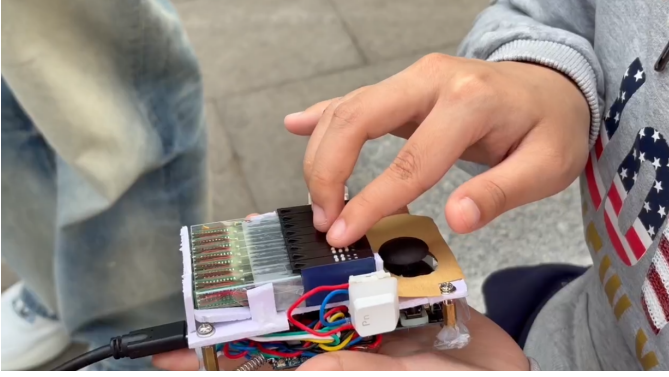


Fig. 5. Real-world testing of BrailleGuide in an outdoor environment. The user is holding and reading the Braille terminal. (The participant agreed to take photos and use them for research purposes.)

A preliminary outdoor evaluation was conducted with a visually impaired participant. The participant successfully captured images and received VLM-generated environmental descriptions through the Braille terminal, reporting that the output was clear and useful. The participant noted that response time could be improved and that the Braille stream sometimes split Chinese characters, which require about 2 to 4 cells each.

This test was designed to assess prototype feasibility: whether the complete interaction chain could operate in a real outdoor setting and produce interpretable Braille feedback. It was not a comprehensive test of user value, accessibility benefit, clinical effectiveness, or generalization, because the evaluation involved one participant, limited scenarios, and primarily qualitative observations.

This splitting reflects token-level paging; future character-aware grouping will keep each Chinese character's complete 2–4-cell Braille sequence together across page boundaries.

VII. DISCUSSION

BrailleGuide explores a tactile-first interaction path for VLM-based assistive perception. Compared with audio-only systems, Braille feedback can provide private, persistent, and rereadable information, which is useful in quiet, noisy, or privacy-sensitive environments. However, the limited number of Braille cells also requires the system to compress visual information into concise and directional descriptions. Therefore, the current design emphasizes structured output, field-based paging, and user-controlled requests instead of continuous long-form scene narration.

Future versions of BrailleGuide can be extended with more input and output modalities. Audio can be added as an auxiliary input channel, allowing users to ask natural-language questions when button or joystick input is insufficient. Audio can also be added as an auxiliary output channel for urgent warnings, short confirmations, or optional summaries. In this direction, audio should complement rather than replace Braille feedback: Braille remains the primary channel for reviewable

environmental information, while speech and sound cues are used only when they reduce interaction cost or improve situational awareness.

VIII. CONCLUSION

BrailleGuide, proposed in this research, combines VLM-based scene understanding with refreshable Braille feedback in a split wearable assistive system. The prototype and preliminary test suggest that it can provide concise and useful environmental descriptions for visually impaired users.

The main outcome of this work is a feasible end-to-end prototype that links image capture, VLM-based environmental interpretation, structured response processing, and user-controlled Braille interaction. Future work should evaluate the system with more participants and repeated scenarios, using quantitative measures such as response latency, field-selection accuracy, reading burden, task completion, and perceived usefulness before stronger claims about practical value can be made.

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